Workshop

Building and Accessing Cortex Agents with Google Cloud & Snowflake

Task 0. Setup and Requirements

Before you click the Start Lab button

Read these instructions. Labs are timed and you cannot pause them. The timer, which starts when you click **Start Lab**, shows how long Google Cloud resources will be made available to you.

This hands-on lab lets you do the lab activities yourself in a real cloud environment, not in a simulation or demo environment. It does so by giving you new, temporary credentials that you use to sign in and access Google Cloud for the duration of the lab.

What you need

To complete this lab, you need:

- Laptop and access to a standard internet browser (Chrome browser recommended).
- Time to complete the lab.

Note: If you already have your own personal Google Cloud account or project, **do not use it** for this lab.

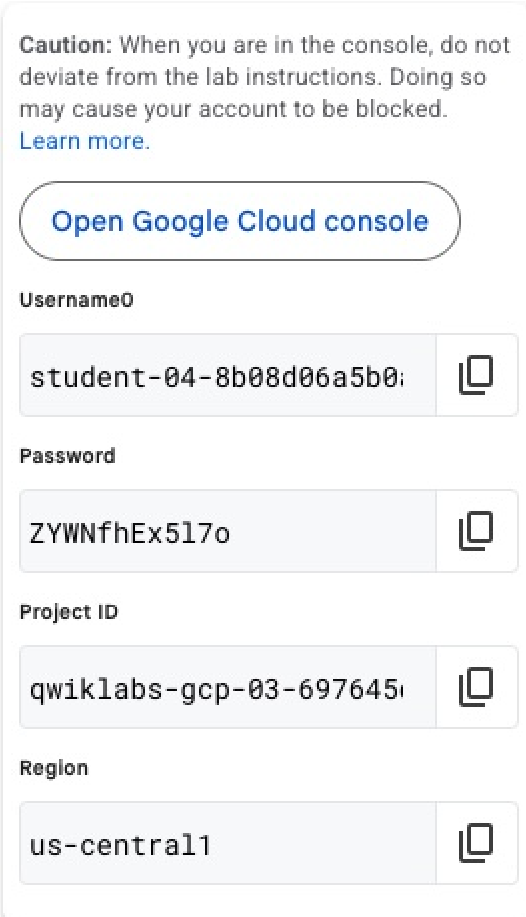
Note: If you are using a Chrome OS device, open an Incognito window to run this lab.

Alternatively, you can log out of all other Google / Gmail accounts before beginning the labs.

How to start your lab and sign in to the Google Cloud Console

Tip: When prompted, select the option to use a new Chrome profile to keep this lab separate from other Chrome activities.

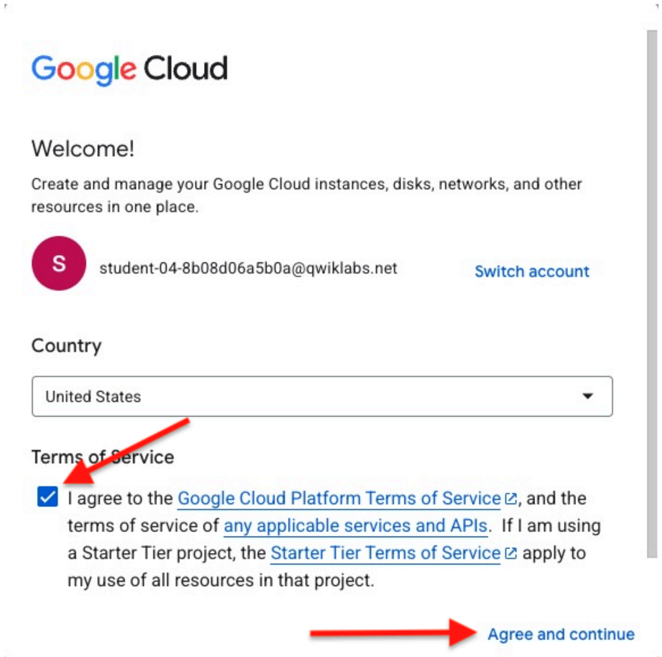
1. Click the **Start Lab** button. On the left is a panel populated with the temporary credentials that you must use for this lab.



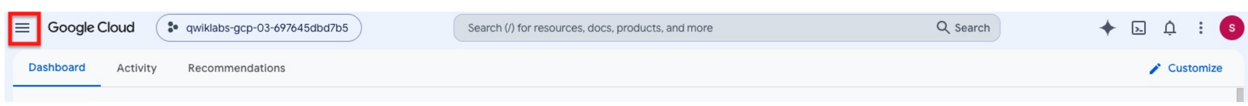
2. Copy the username, and then *right-click* **Open Google Console** and select **Open Link in Incognito Window.** The lab spins up resources, and then opens the incognito window. You'll be prompted to read and acknowledge the Welcome to Your New Account details. Click "I Understand".

   You'll be presented with the Terms and Conditions. Verify that your account matches the Username from your lab credentials. Click the checkbox to agree to the Terms of Service, and then click on "Agree and Continue".

   Important: You must use the credentials provided for the lab. Do not use your own Google Cloud Training credentials. If you have your own Google Cloud account, do not use it for this lab (avoids incurring charges).

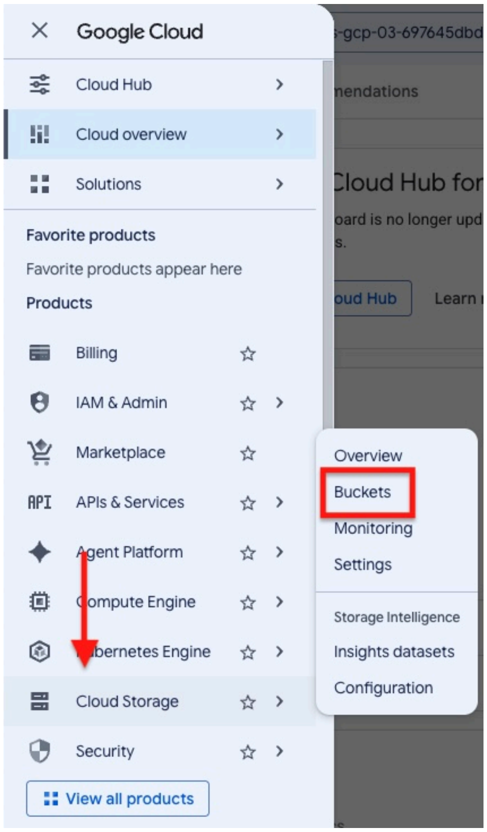


3. After a few moments, the Google Cloud will open.
   Note: You can view the menu with a list of available Google Cloud Products and Services by clicking the **Hamburger menu** at the top-left.

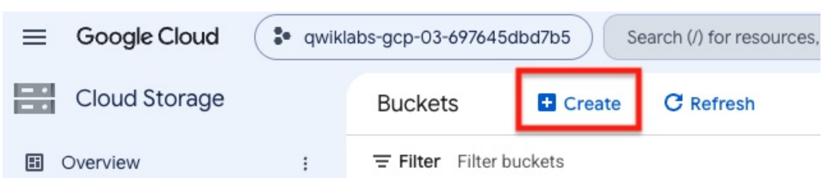


Task 1. Setup Your Google Cloud Storage Bucket

1. We will create a Google Cloud Storage bucket to hold the data for the Iceberg table that we will be creating in Snowflake. In the left navigation panel, hover over **Cloud Storage** and select **Buckets**.

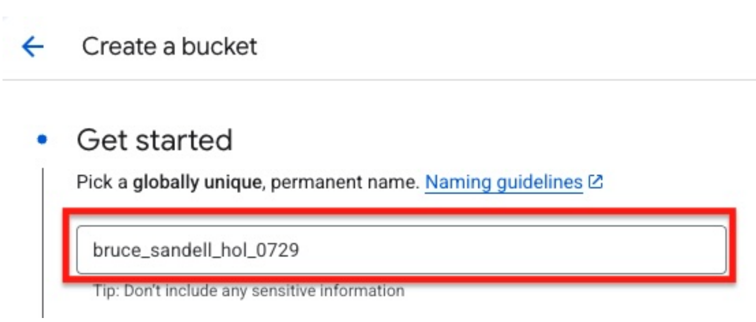


2. Click the “Create” link to start the bucket creation process.

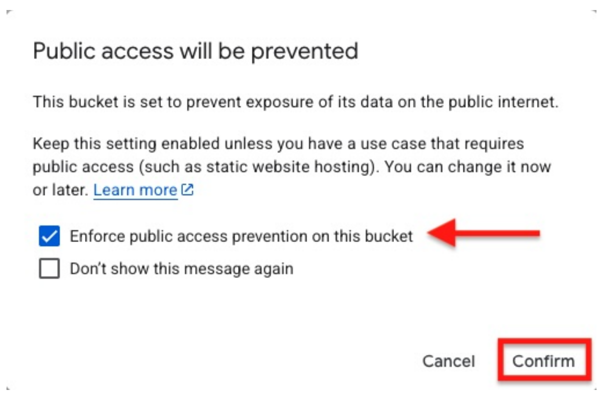


3. Add your bucket name as *firstname_lastname_hol_0729*. (Note, keep this bucket name copied and pasted into a notepad/textpad as we will need it later.)

   We’ll be using all of the default values for creating the bucket, so you can scroll all the way to the bottom of the page and click “Create”.

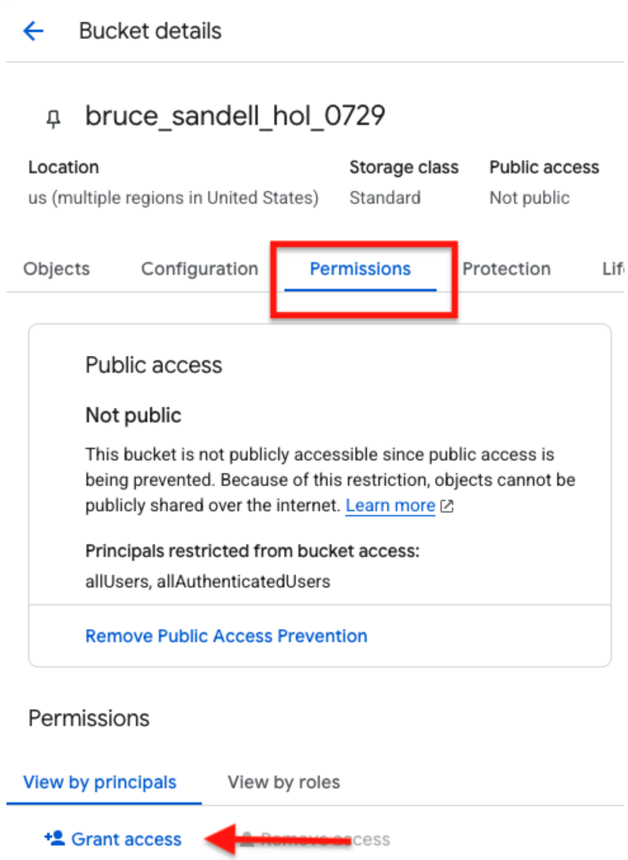


4. Leave **Enforce public prevention** checked and click **Confirm**.

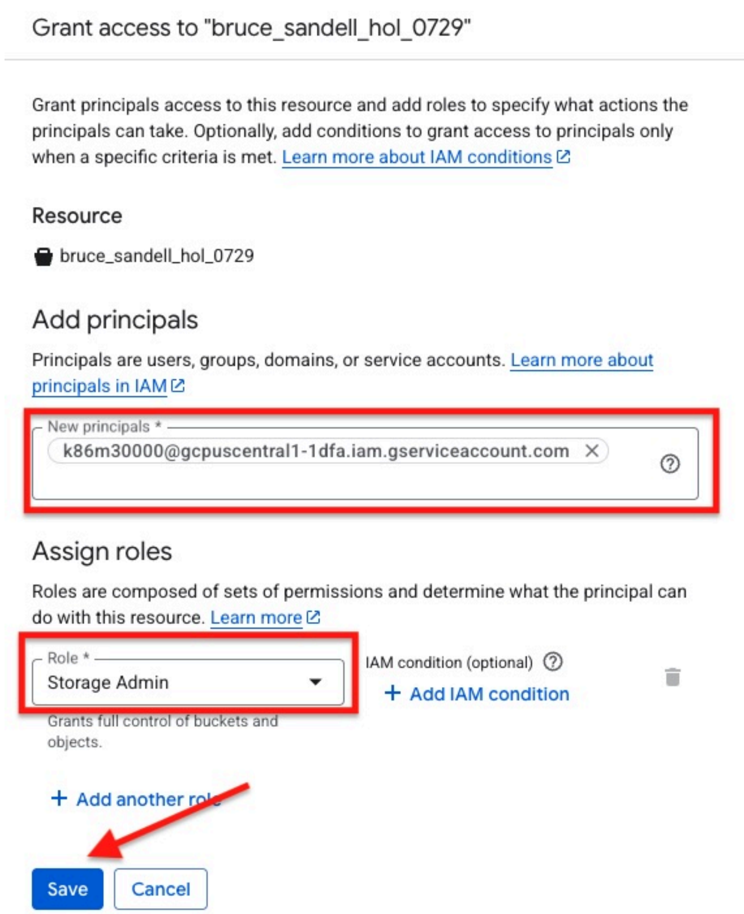


5. **Return to your Snowflake Notebook**. Once you have completed the step where you CREATE EXTERNAL VOLUME, return here with the value for STORAGE_GCP_SERVICE_ACCOUNT as described in the Notebook.

6. Click on the **Permissions** tab on the bucket screen so we can add permissions for the Snowflake service account. Then click the **Grant Access** link.



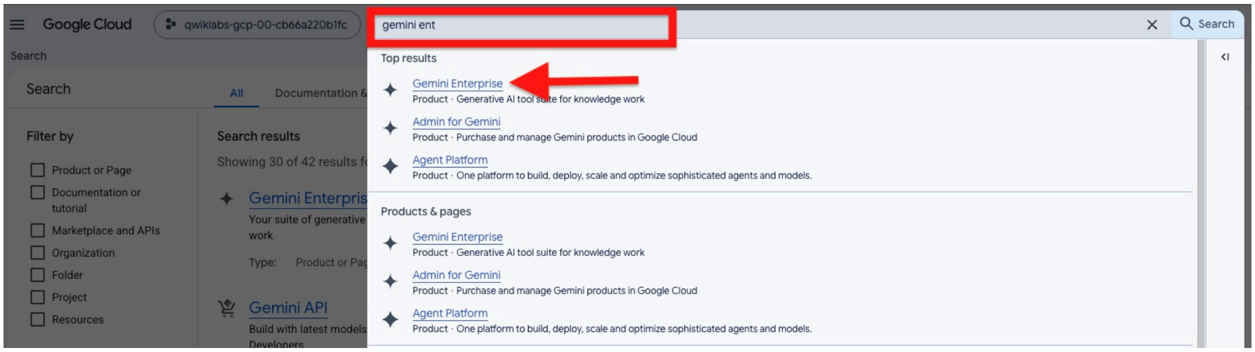
7. On the **Grant Access** screen, paste the Snowflake STORAGE_GCP_SERVICE_ACCOUNT (which would have been returned when you ran the SQL command "*DESC EXTERNAL VOLUME hol_gcs_vol;*" in Snowsight) into the **New principals** box, and click enter. Under **Assigned roles**, click inside to open the filter box. In the filter enter **storage admin**, and select the **Storage Admin** entry. Then click **Save**.



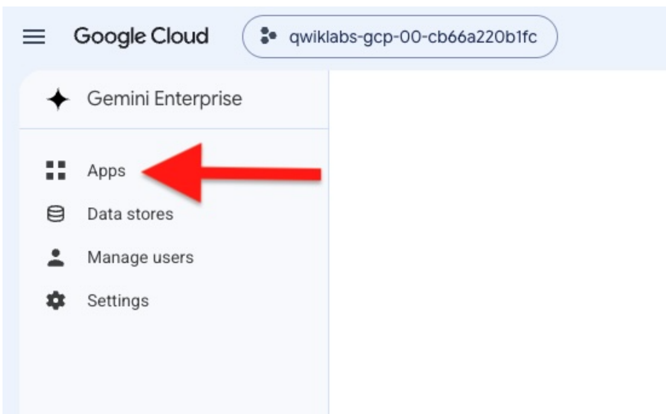
Task 2. Create your Custom MCP Server Data Source

1. Before starting this step, please ensure that you have completed the creation of your MCP Server in the Snowflake Lab instructions.

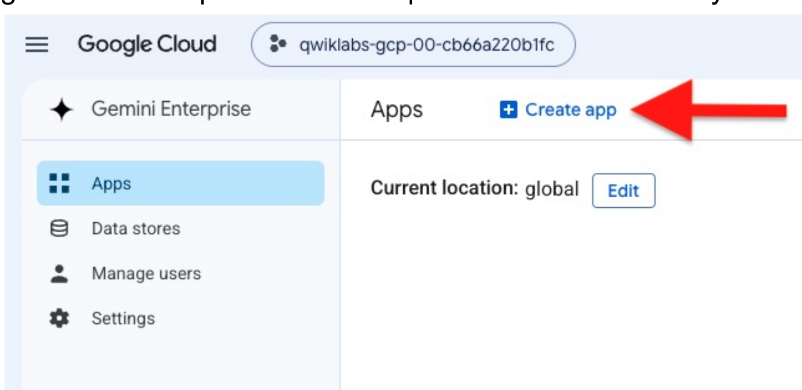
2. In the Search Field at the top of the Google Cloud Console, enter **Gemini Enterprise** and then click on Gemini Enterprise from the list of products displayed to go to the Gemini Enterprise Home Screen.



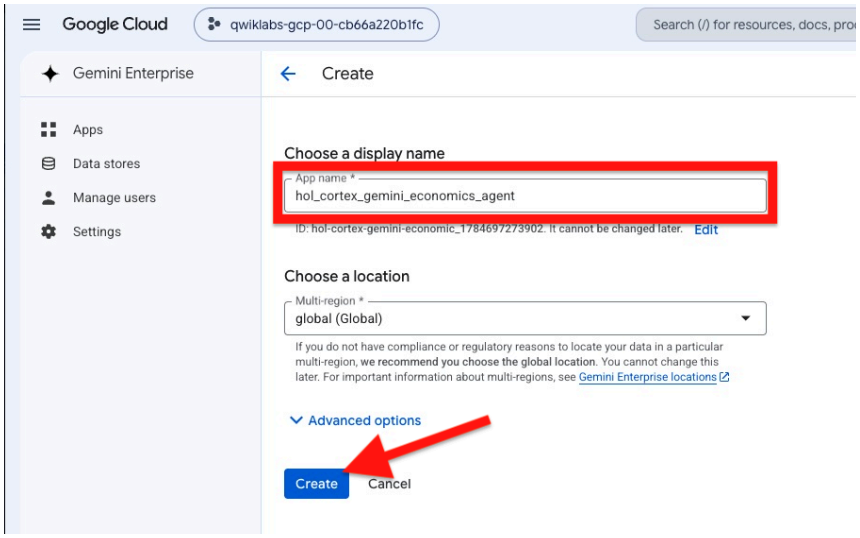
3. From the Gemini Enterprise Home Screen, Click on **Apps** in the left-side panel.



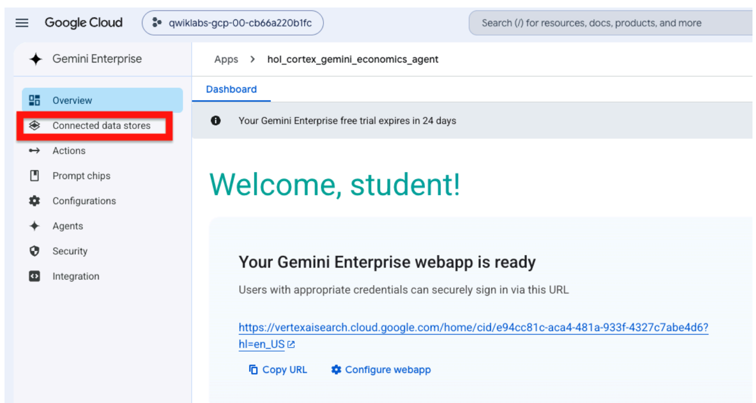
4. Select “Create App”. A Gemini Enterprise App is a centralized, AI-powered workplace platform designed for teams to collaborate, automate workflows, and deploy custom AI agents without writing code. It acts as a unified hub where employees can access grounded enterprise data and specialized tools securely.



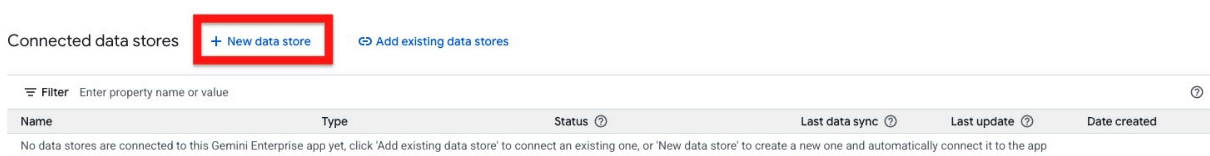
5. Choose a name for your Gemini Enterprise App. You can use the one suggested in the Notebook, or create your own. Press “Create” to create your new app. It will take a few seconds to create the new app.



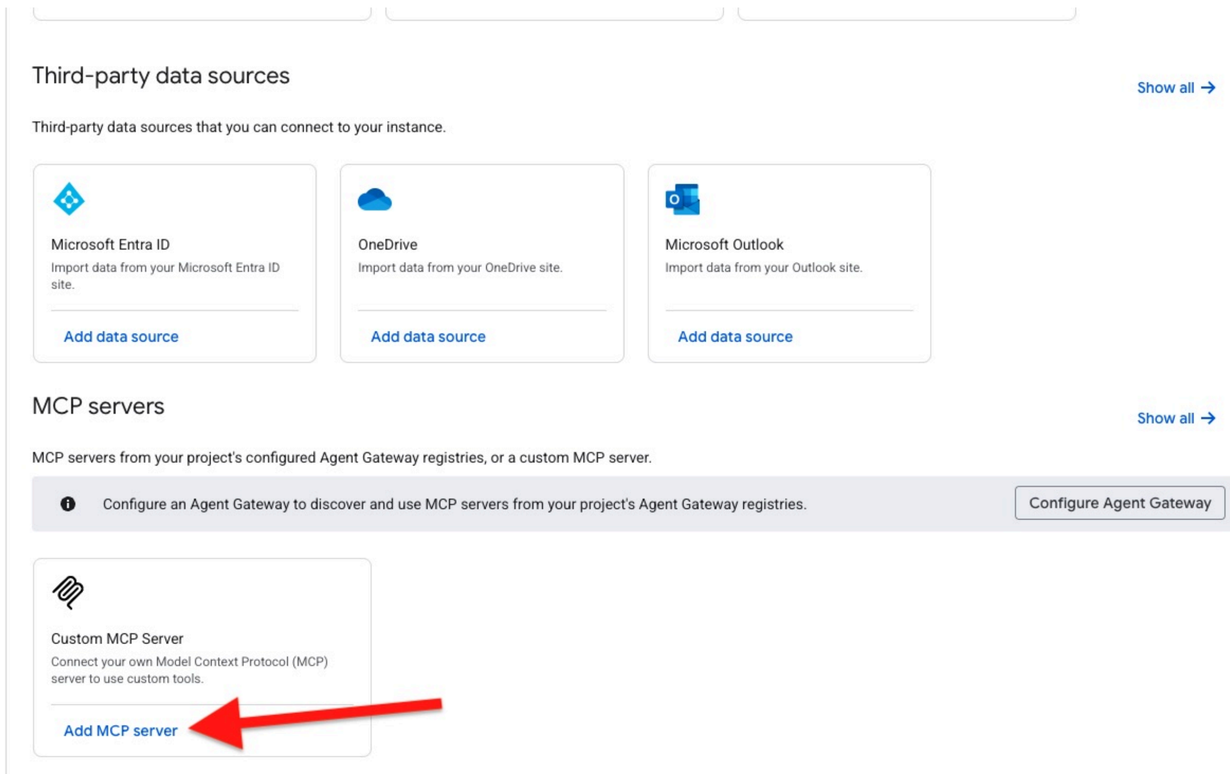
6. From the Apps home page, click on “Connected data stores”



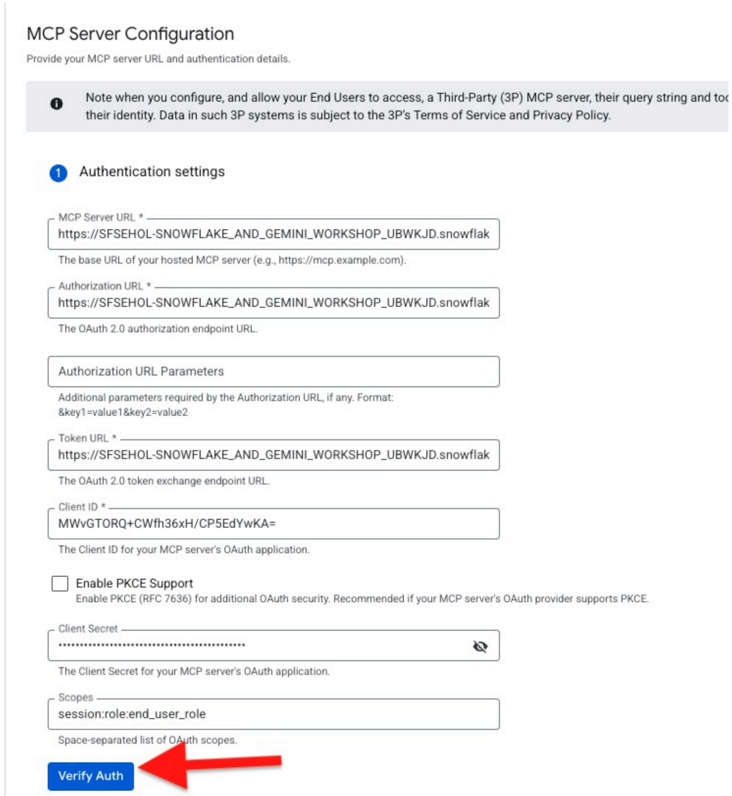
7. Click on “New Data Store”



8. Scroll through the list of available data stores until you get to the “MCP Servers” section. In the data source named “Custom MCP Server”, click on “Add MCP server”.

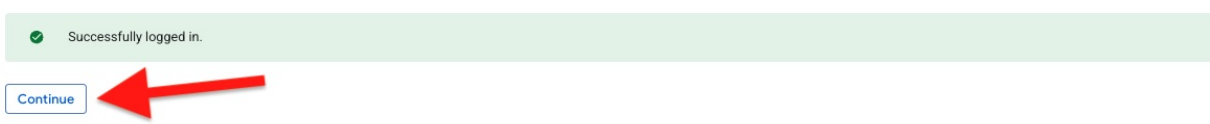


9. Leave OAuth 2.0 selected as the verification method. Populate the fields with the data supplied by the Snowflake Notebook. Every field should be populated except “Authorization URL Parameters”.



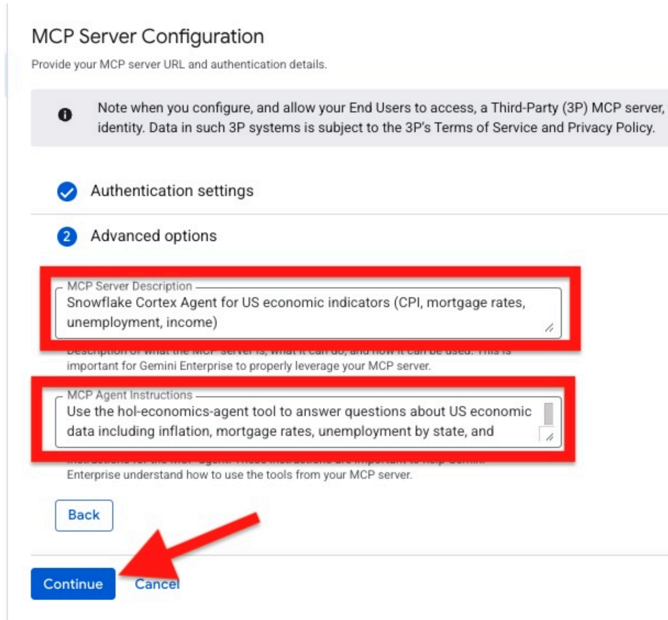
Click " Verify Auth" to go through the Snowflake OAuth process and ensure that all of your authentication details are set up correctly. Use the same Username and password that you used when you initially logged into Snowflake (username = USER, Password =). If you are prompted to allow access to your account, click "Allow".

10. After successfully authenticating, click on "Continue".

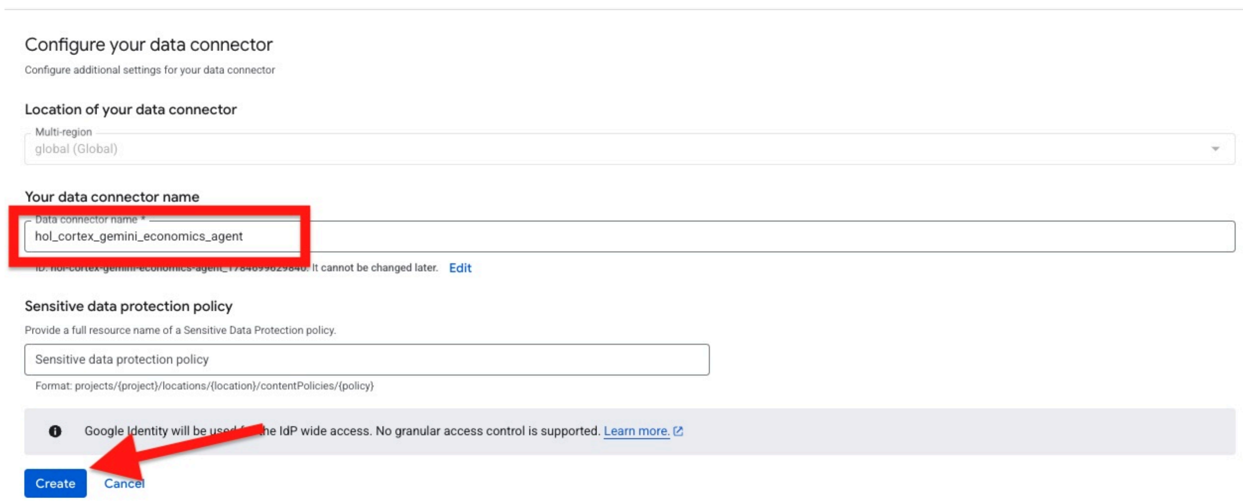


11. In the "Advanced Options" section, populate the MCP Server Description and MCP Agent instructions. Suggested values for these fields should have been supplied in your

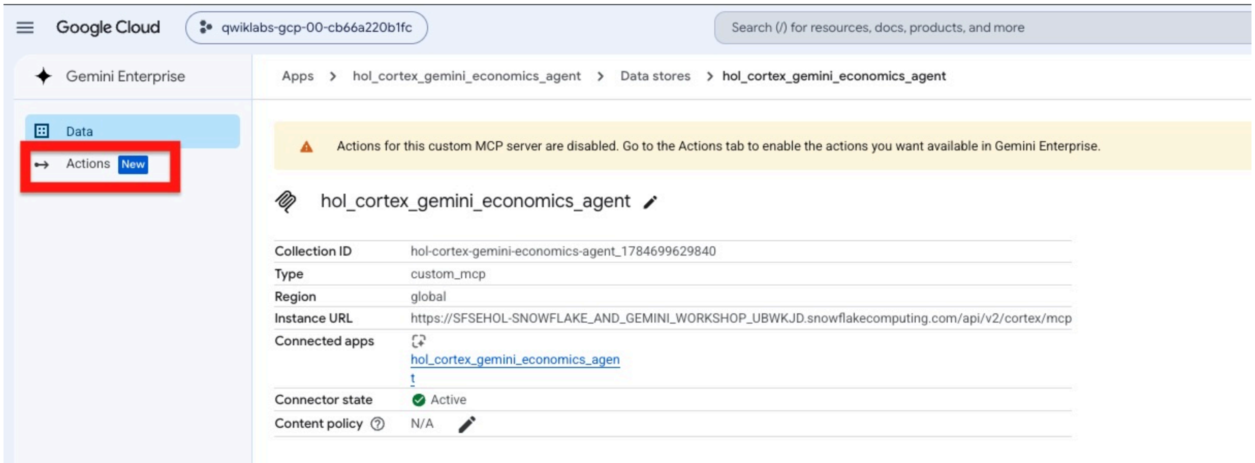
Snowflake Notebook. Click “Continue”.



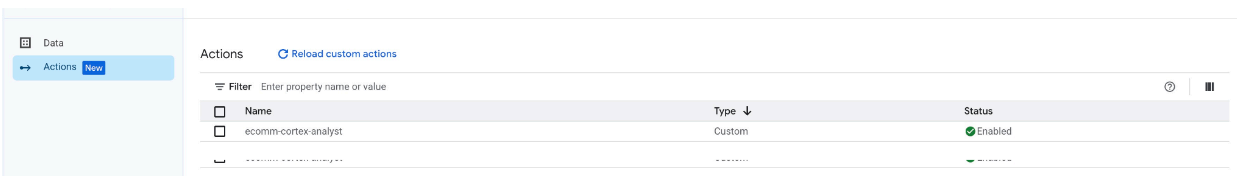
12. Provide a name for your new data connector.   You can use the one supplied by the Snowflake Notebook, or make up your own name. Leave the other fields with their default values. Click “Create”.



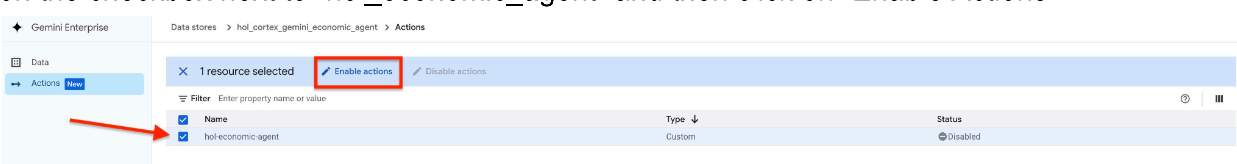
13. It may take several seconds for the data source to be created. Once the data source has been created, you will be returned to the Home screen for your new data source. In the menu on the left, click “Actions”.



14. MCP Servers may expose one or more available actions. Click "Reload custom actions" to get a list of all of the available actions. You will be prompted to go through the OAuth process again. Use the same Username and Password that you used for the earlier authentication step. For this lab, there should only be one available action.



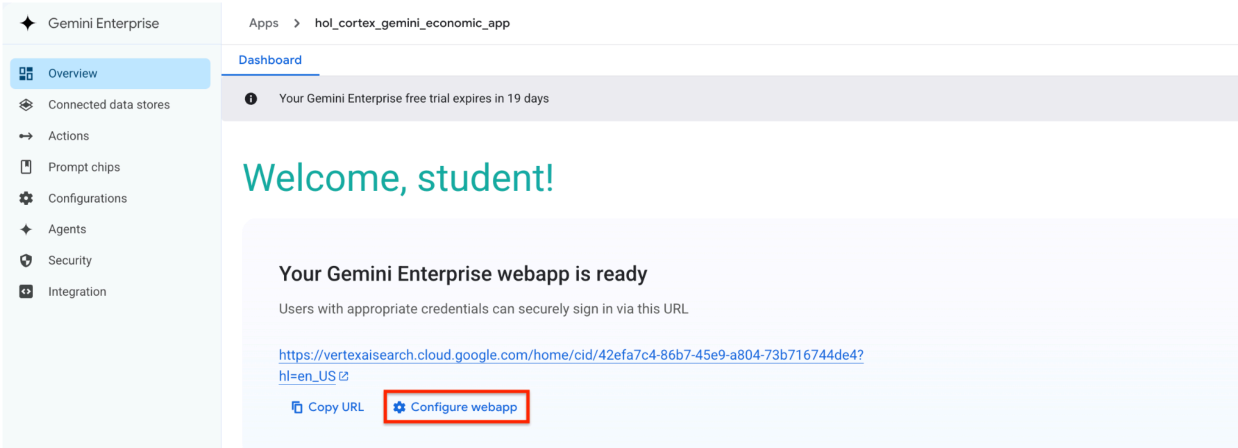
15. You now have the option to select which of the available actions that you want to expose to your end users. Since there is only one action available, we want to enable it. Click on the checkbox next to "hol_economic_agent" and then click on "Enable Actions"



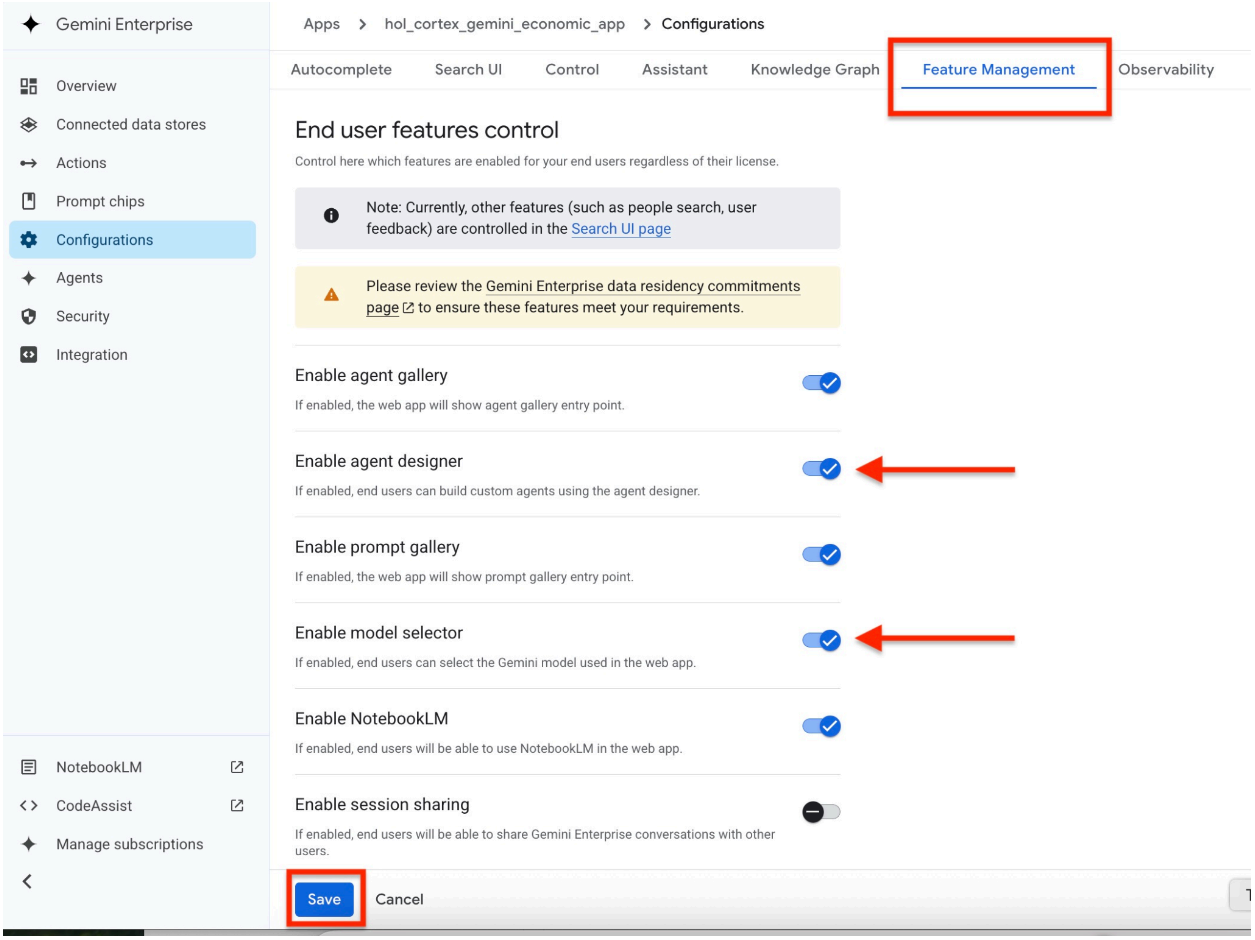
The "Disabled" Indicator should change to "Enabled".

Task 3. Configure the Gemini Enterprise Webapp

1. Your new app should be ready to customize. Before users start accessing the webapp, you can configure what features will be exposed to them. Click on "Configure webapp" to see all of the configuration options.



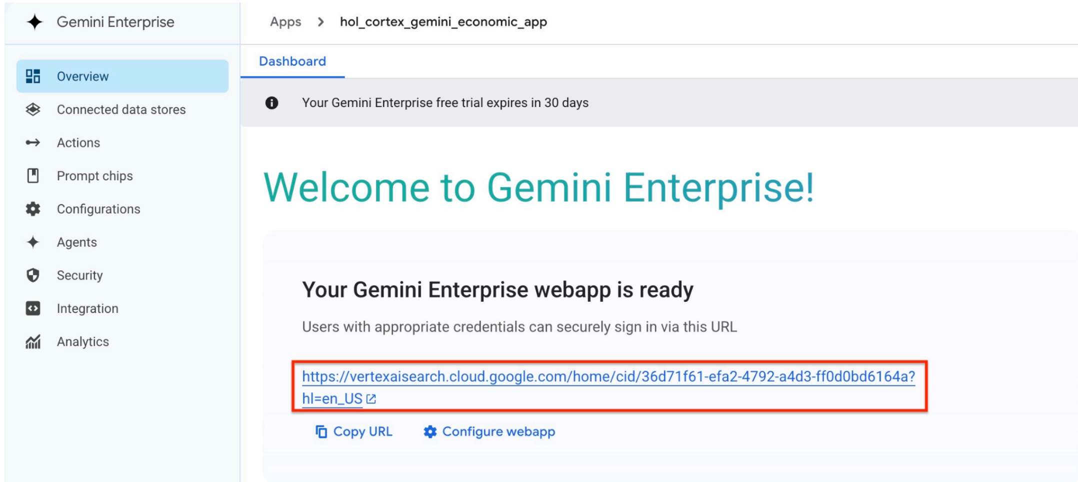
2. Review the features that are available to configure by clicking through the tabs at the top of the page. For this lab, there are two features that we are going to enable. Click on the “Feature Management” tab and enable the features:
   Enable agent Designer - allows users to create their own agents using graphical tools
   Enable model selector - allows users to pick the most appropriate model for their agent.



3. Navigate back to the app Home page by clicking the “Overview” tab on the left side menu.

Task 4. Work with your new Gemini Enterprise Webapp

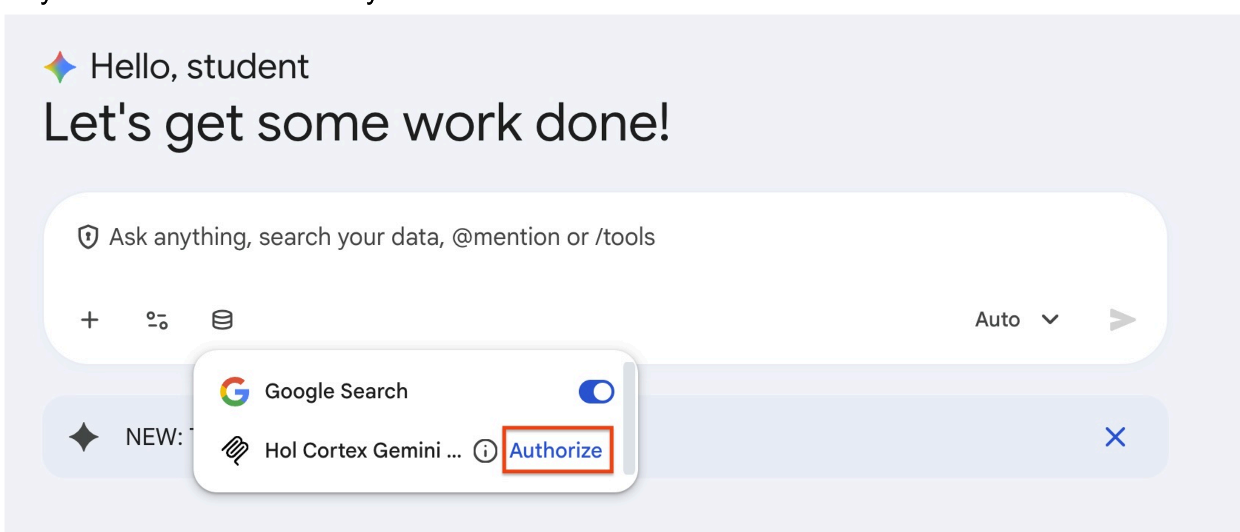
1. Now that the Webapp has been configured, it’s time to try it out. Open the Webapp by clicking the link on the App Overview screen.



2. Once the Webapp is open, take a moment to click through the options on the bottom of the "Ask Anything" data entry field. You can add files for searching, enable tools and specify which connectors to use.

3. Click on the disk icon in the data entry field and click "Authorize" to enable the connection to the Snowflake MCP server. You'll need to authenticate to Snowflake using the same credentials that you have used in previous steps.

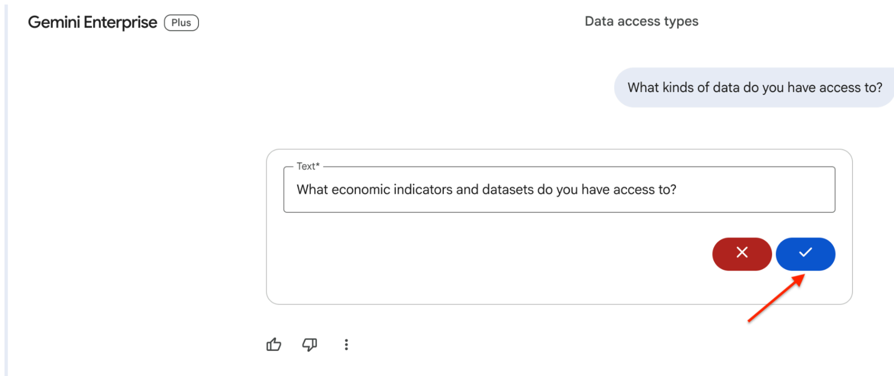
   Once the authentication has completed successfully, disable Google Search, so that the only enabled data source is your Snowflake MCP Server.



4. Enter a question into the data entry form. A good place to start is: "What kind of data do you have access to?"

   Before the question is submitted, you may be requested to accept the prompt that

Gemini is planning to send to your Snowflake agent. You can accept the prompt by clicking on the blue check mark button. You are also free to rate the quality of the generated prompt using the "Thumbs Up" or "Thumbs Down" icon, which will help the Gemini model learn. You can modify the prompt before it is sent as well.



5. You will be prompted with some suggested questions, or you can come up with your own. You can have a multi-turn conversation with the data, meaning that the context of the previous question is used as the input to the followup question. For example:
    - What was the annual unemployment rate in the United States for 2025?
    - Can you break that down by month?
    - Can you break down the January data by state?

    When you are ready to move on to a new topic, click the "New Chat" link on the left menu. Each time that you open a new chat link, Google Search will be enabled by default, so you may want to disable it.

6. Continue exploring the data!